

BUREAU OF INDIAN STANDARDS

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भारतीय मानक मसौदा
फर्नीचर के लिए अपहोलस्टरी
चमड़ा — विशिष्टि

(IS 577 का दूसरा पुनरीक्षण)

Draft Indian Standard
**Upholstery Leather for
Furniture — Specification**
(Second Revision of IS 577)

(ICS 59.140.30)

Leather, Tanning Materials and Allied Products Sectional Committee, CHD 17

Last Date for Comments: As per covering letter

FOREWORD

(Formal clause to be added later)

Leather is extensively used to make myriad products namely bags, purses, clothes, upholstery material, shoes, belts, interior material or wallpapers etc. Due to its uniqueness as a natural material for use in different weather conditions like cool in summer and warm during winter, besides durability and versatility, leather is still considered as supreme alternative to many other covering materials and upholstery is no exception. The upholstery leather industry, like many other branches of the industry, is in a rapid state of progress, largely brought about by the ever-increasing demands of the fashion conscious consumer and the acute competition from synthetics. Leather continues to be the ideal upholstery material due to its unique surface and other functional characteristics.

This standard was originally published in 1954. The first revision was published in 1986. During the first revision the following modification had been incorporated:

- Requirements on oils and fats content, differential number, tightness of grain, crackiness of grain, tackiness and cracking had been deleted;
- Requirements on fastness properties, blocking resistance, flexural endurance and adhesion of finish had been incorporated;

c) Three grades of upholstery leather based on the end use had been prescribed. Grade I is recommended for furniture without upholstered arms and for gentle use. Grade 2 is recommended for furniture for general usage and Grade 3 is recommended for furniture subjected to severe weathering conditions.

During this second revision the following modification have been incorporated:

- a) Requirements for hazardous chemical has been added (*see* Table 2) as these chemicals may pose health risks to the users in direct contact with the leather, besides affecting the environment on disposal;
- b) Requirements on elongation at break, tear strength, and stitch tear strength has been added;
- c) Grades have been merged to harmonize with ISO 16131; and
- d) Title of the standard has been revised as “Upholstery Leather for Furniture — Specification”

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2: 2022 ‘Rules for rounding off numerical values (*second revision*)’. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

1 SCOPE

This standard prescribes the requirements, methods of sampling, and tests for upholstery leather for furniture.

2 REFERENCES

The Indian standards listed in Annex A contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this Indian standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

3 TERMINOLOGY

3.1 For the purpose of this standard, the definition given in IS 1640 and the following shall apply:

3.1.1 *Blocking*

Unintentional adhesion between the finished surfaces of leather or between the finished surface of a leather and another surface.

4 REQUIREMENTS

4.1 Raw Materials

The material shall be dry, dry-salted, or wet-salted hides in the form of evenly trimmed whole hides. The area of the hide is approximately 2 square metres or more, and sizes other than this are as agreed between the purchaser and supplier. It shall have at most four cuts and holes, which shall not restrict the selection of usable panels from the piece of leather at a distance more than 250 mm from the edges. The cuts may be either deep through or surface cuts, but the length should not exceed 25 mm in case of longitudinal cuts and 25 mm in diameter in case of round holes. The nature of defects may include butcher cuts, pin or sore holes, healed-up wounds, patches, and scratches. Scratches due to thorns, barbed wire, or horn shall be more than 50 mm long. A few widely scattered pox marks may be allowed.

4.2 Tanning

4.2.1 The leather may be chrome tanned or combination tanned or other tanning as agreed between the purchaser and supplier.

4.2.2 *Treatment with Preservatives*

At the end of the tanning process and before drying, the leather shall be treated with suitable biocides in requisite proportions to promote mildew resistance. The material shall show no growth of mildew when examined visually.

The chosen biocides shall be effective against microorganisms, non-toxic, or within the prescribed limits as specified in Table 2.

4.3 Dyeing

The leather shall be dyed to an even shade, and the dye shall penetrate throughout the grain layer or as agreed between the purchaser and supplier.

4.4 Finishing

The leathers shall be soft and pliable and evenly finished or as agreed between the purchaser and supplier.

4.5 Physical and chemical requirements

The material shall comply with the physical requirements as specified in Tables 1 and chemical requirements as specified in Table 2, when tested in accordance with the methods prescribed in the Table 1 and Table 2.

Table 1 Physical requirement for Leather Upholstery

(Clause 5.5)

Sl No.	Characteristics	Requirements	Test method, Ref To
(1)	(2)	(3)	(4)
i)	Thickness, mm	As agreed between the purchaser and supplier, with a tolerance of 0.1 mm	LP:1 of IS 5914
ii)	Tensile strength, N/mm ² , <i>Min</i>	10	IS 5914 (Part 8)
iii)	Elongation at Break, percent	35 to 75	
iv)	Tear strength (Double Edge), N, <i>Min</i>	20	IS 5914 (Part 5/Sec 2)
v)	Stitch Tear Strength, N/mm, <i>Min</i>	40	LP : 8 of IS 5914
vi)	a) Colour fastness to rubbing (Grey Scale Rate)	Staining/Marring	IS 6191 (Part 4)
	b) Coated, Pigment and others finish	Grade 4 Grade 4	
	1. Dry 500 rubs, <i>Min</i>		
	2. Wet 250 rubs, <i>Min</i>	Grade 3 Grade 3	
	c) Nubuck and Aniline finish		
	1. Dry 50 rubs, <i>Min</i>		
	2. Wet 20 rubs, <i>Min</i>		
vii)	Colour fastness to water and perspiration, <i>Min</i> (Gray Scale Rate)	Staining Grade 3/4	IS 6191 (Part 2)
			IS 6191 (Part 6)
viii)	Colour fastness to water spotting		IS 6191 (Part 1)
	a) For 30 min	No change or permanent mark	

	b) For 16 h	No change or permanent mark	
	c) Grey Scale Rate, <i>Min</i>	Grade 4/5	
ix)	Flexing Endurance	No crack	IS 5914 (Part 6/Sec 1)
	a) Coated, Pigment and others finish Dry flexes: 50 000		
	b) Aniline finish Dry flexes: 20 000		
x)	Adhesion of finish, N/cm		IS 6191 (Part 5)
	a) Coated, Pigment and others finish		
	1. Dry, <i>Min</i>	3	
	2. Wet, <i>Min</i>	2	
	b) For Patent finish		
	1. Dry, <i>Min</i>	5	
	2. Wet, <i>Min</i>	3	
xi)	Abrasion resistance (Martindale)	Not worse than slight wear	IS 8085 (Part 6)
	a) Dry: 51 200 cycles		
	b) Wet: 25 600 cycles		
xi)	Colour fastness to light	Marring	IS/ISO 105 B02
	a) Grey Scale Rate		
	b) Coated, Pigment and others finish, Grade, <i>Min</i>	5	
	c) Nubuck and Aniline finish, Grade, <i>Min</i>	4	
xii)	Colour fastness to migration (Applicable for light colour leathers) Grey Scale Rate, <i>Min</i>	4/5	IS 6191 (Part 7)
xiii)	Burning Behavior, mm/min, <i>Max</i>	80	ISO 3795
xiv)	Softness, Deflection, mm, <i>Min</i>	3	IS 5914 (Part 13)

Table 2 Compliance with Hazardous Chemicals

(Clause 5.5)

Sl No.	Critical Chemical Substances	Requirements	Test method, Ref To
(1)	(2)	(3)	(4)
i)	Aromatic amines of azo dyes (individual or sum of the listed amines), mg/kg, <i>Max</i>	30.0	IS 582 (Part 5/Sec 1) or IS 582 (Part 5/Sec 2)
ii)	Chlorophenols (individual or sum of the listed chlorophenol variants, including isomers), mg/kg, <i>Max</i>	5.0	IS 14575
iii)	Formaldehyde, mg/kg, <i>Max</i>	Adults: 150.0 Children: 75.0	IS 16297 (Part 1) or IS 16297 (Part 2)
iv)	Chromium VI, mg/kg, <i>Max</i>	3.0	IS 582 (Part 11/Sec 1) or IS 582 (Part 11/Sec 2)
v)	Preservatives		IS 582 (Part 8/Sec1) or IS 582 (Part 8/Sec2)
	a) 2-(thiocyanomethylthio)-benzothiazole, (TCMTB), mg/kg, <i>Max</i>	300	
	b) 4-chloro-3-methylphenol, (PCMC), mg/kg, <i>Max</i>	300	
	c) 2-phenylphenol, (OPP) mg/kg, <i>Max</i>	750	
	d) 2-octylisothiazol-3(2H)-one, (OIT), mg/kg, <i>Max</i>	100	
	e) Phenol, mg/kg, <i>Max</i>	100	
vi)	Alkylphenolethoxylates		IS 582 (Part 15/Sec 1) or IS 582 (Part 15/Sec 2)
	a) Octylphenolethoxylate, mg/kg, <i>Max</i>	500	
	b) Nonylphenolethoxylate, mg/kg, <i>Max</i>	500	

6 MARKING AND PACKING

6.1 Marking

6.1.1 The following information shall be clearly and permanently marked on flesh side of leather;

- Area of leather;
- Manufacturer's identification mark;
- Month and year of the tanning and finish coat; and
- Reference to this international standard.

6.1.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

6.2 Packing

6.1 The upholstery leathers shall be packaged suitably as agreed between the purchaser and supplier in such a way that it is protected from humidity, heat, light etc.

7 Sampling

Sampling of leather shall be as specified in IS 5868 (Part 1). The location of test pieces shall be as specified in IS 5868 (Part 2). The conditioning of test pieces shall be as specified in IS 5868 (Part 3).

ANNEX A

(Clause 2)

LIST OF REFERRED INDIAN STANDARD

IS	Title
IS/ISO 105 (Part B02):2014	Textiles – Tests for colour fastness – Part B02 Colour fastness to artificial light: Xenon arc fading lamp test
IS 582	Methods of chemical testing of leather (<i>first revision</i>)
Part 5	Determination of certain azo colorants in dyed leathers
Sec 1: 2018/ ISO 17234-1: 2015	Determination of certain aromatic amines derived from azo colorants
Sec 2:2018/ ISO 17234-2:2011	Determination of 4-Aminoazobenzene
Part 8	Determination of the Preservative (TCMTB, PCMC, OPP, OIT) Content in Leather by Liquid Chromatography
Sec 1: 2024/ ISO 13365-1: 2020	Acetonitrile Extraction Method
Sec 2: 2024/ ISO 13365-2: 2020	Artificial Perspiration Extraction Method
Part 11	Determination of chromium (VI) content in leather
Sec 1:2022/ ISO 17075-1	Colorimetric method
Sec 2:2022/ ISO 17075-2	Chromatographic method
Part 15	Leather determination of ethoxylated alkylphenols APEO
Sec 1: 2024/ ISO 18218-1: 2023	Direct Method
Sec2: 2024/ ISO 18218-2: 2019	Indirect Method
IS 1640: 2025	Hides, Skins and Leather — Vocabulary (<i>second revision</i>)
IS/ISO 3795:1989	Road Vehicles and Tractors and Machinery for Agriculture and Forestry - Determination of Burning Behaviour of Interior Materials
IS 5868	Leather - Methods of Sampling
Part 1: 2025	Number of items for a gross sample (<i>second revision</i>)
Part 2: 2025	Position and Preparation of Specimens for Testing for Chemical, Physical, Mechanical and Fastness Tests (<i>second revision</i>)
Part 3: 2026	Specimen and Test Piece Conditioning for Physical and Mechanical Tests (<i>second revision</i>)
IS 5914	Methods of physical testing of leather
LP:1	Method for measurement of thickness of test pieces
LP:8	Determination of double hole stitch tear strength
Part 5	Determination of tear load
Sec 2:2023 ISO 3377-2:2016	Double edge tear
Part 6	Determination of flex resistance
Sec 1:2023 ISO 5402-1 : 2022	Flexometer method
Part 8: 2023/ ISO 3376: 2020	Determination of tensile strength and percentage elongation
Part 13: 2025/ ISO 17235: 2015	Determination of softness
IS 6191	Micro - biological colour fastness and microscopical tests for leather
Part 1:2018/ ISO 15700: 1998	Colour fastness to water spotting
Part 2: 2017/ ISO 11642: 2012	Colour fastness to water
Part 4: 2018/	Colour fastness to cycles of to-and-fro rubbing (<i>first revision</i>)

ISO 11640: 2012	
Part 5: 2024/ ISO 11644: 2022	Test for adhesion of finish (<i>first revision</i>)
Part 6: 2024/ ISO 11644: 2022	Test for adhesion of finish (<i>first revision</i>)
Part 7: 2025/ ISO 15701: 2022	Colour fastness to migration into polymeric material
IS 8085 (Part 6): 2021/ ISO 17704: 2004	Methods of test for footwear Part 6 abrasion resistance of uppers, linings and insoles
IS 14575: 2024/ ISO 17070:2015	Leather — chemical tests — determination of tetrachlorophenol - trichlorophenol-, dichlorophenol-, monochlorophenol- isomers and pentachlorophenol content (<i>first revision</i>)
IS 16297	Leather chemical determination of formaldehyde content
Part 1: 2014/ ISO 17226-1: 2008	Method using high performance liquid chromatography
Part 2: 2022/ ISO 17226-2: 2018	Method Using Colorimetric Analysis (<i>first revision</i>)